

CampusFlow — Scenario Briefing

Software Engineering and Project Management | FHWN, Bachelor Computer Science, 2nd Semester

The Project

A mid-sized university needs a new system to manage its core academic operations. The current landscape is fragmented: enrollment runs through a legacy Student Information System, grades are tracked in spreadsheets and isolated databases, and scheduling relies on manual coordination between departments. The result is data inconsistencies, slow processes, and frustrated users across the board.

CampusFlow is the proposed solution — a modern, integrated platform that unifies enrollment, scheduling, and grading into a single system. Your team has been brought on as the engineering team responsible for building it.

Core Capabilities

Area	What it covers	—	—	Enrollment	Student applications, course registration, prerequisite enforcement, waitlist management
				Scheduling	Course offerings, room assignments, conflict detection, calendar views
				Grading	Grade submission by faculty, transcript generation, GPA calculation

Who Uses CampusFlow

Stakeholder	Primary interest	—	—	Students	Register for courses, view schedules, access grades and transcripts
				Faculty	Manage course details, submit grades, enforce academic prerequisites
				Administrative Staff	Process enrollments, maintain records, handle exceptions and policy enforcement
				Registrar Office	Certify official records, ensure data integrity across systems

Other parties are affected by CampusFlow but may not interact with it directly. Identifying them is part of your work.

Known Constraints

- **Legacy integration.** The existing Student Information System cannot be replaced overnight. CampusFlow must coexist with it during a transition period.
 - **External dependencies.** CampusFlow does not handle payments (Finance system), user authentication (central Identity Provider), or visa verification (International Office systems). It must integrate with these.
 - **Regulatory compliance.** Credit transfer rules, data privacy regulations, and accreditation standards apply. Some processes cannot be fully automated.
 - **Timeline and budget.** Resources are finite. Not everything can be built at once — prioritization decisions will be necessary.
 - **Multi-stakeholder trade-offs.** Students want speed and simplicity. Faculty want control and flexibility. Administrators want consistency and auditability. These goals sometimes conflict.
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What Is Not Defined

This briefing is intentionally incomplete. Among the things you will need to determine:

- Who are all the stakeholders — and what do they actually need?
 - What does a successful enrollment process look like end to end?
 - Where should the system boundary be drawn — what stays manual?
 - Which quality attributes matter most: performance, usability, security, availability?
 - What assumptions are you making — and how would you verify them?
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Your Role

You are the engineering team. Throughout this course, you will analyze the context, identify stakeholders, elicit and specify requirements, design an architecture, plan for testing, and assess readiness — all within the CampusFlow scenario.

Decisions you make early will shape what is possible later. Document your assumptions. Question what seems obvious. Start with the problem, not the solution.

CampusFlow is a fictional scenario designed for educational purposes. Any resemblance to real systems is intentional — the complexity is real, even if the project is not.